

PROJECT SYNOPSIS

1. PROJECT TITLE

Real-Time Computer Vision based American Sign Language to English Translator using MediaPipe for hand landmark detection

2. TEAM DETAILS

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3. INTRODUCTION & BACKGROUND

Sign language acts as a communication bridge between hearing-abled individuals and the deaf/mute community. However, due to the lack of widespread usage and training of sign language among the general public, the differently abled struggle to communicate with others without the help of a dedicated human interpreter. As a result, differently-abled people face greater risk of isolation.

The communication hurdles faced by sign language users are increased when factoring in the various sign languages that exist today. For example, a person fluent in American Sign Language cannot communicate with another person trained in British Sign Language - despite both being "English", they are two distinct languages with their own grammar rules and vocabulary - only sharing about 30% signs with each other. These hurdles point towards a critical need for a translator tool that is accessible and highly reliable for practical day-to-day communication needs.

4. PROBLEM STATEMENT

Existing automated sign language translators generally fall into two categories, both of which possess distinct practical limitations:

1. **Sensor-Based Systems:** These rely on specialized peripheral hardware such as electronic sensor gloves or 3D depth-mapping cameras. Sensor based systems are more structurally precise for detecting hand gestures. However, due to the requirement

of additional hardware, they are cost-prohibitive and impractical for everyday casual communication.

2. **Traditional Computer Vision Systems:** Such systems are more accessible because they use standard cameras which are present in most consumer grade smartphones and desktops of today. However, they are also not without challenges. A vision-based approach is less precise than sensor-based approaches because the vision detection is highly dependent on lighting factors, camera quality, background, etc. Vision-based models also suffer from a severe trade-off between computational efficiency and performance. Lightweight models using basic machine learning handle static gestures (finger-spelling) efficiently but cannot capture the fluid, dynamic gestures that comprise natural conversation. Conversely, heavy deep learning systems built to process raw video files for dynamic recognition require high-end, dedicated GPUs, which tend to be too complex to run on standard laptops or smartphones.

Another limitation present in most hobbyist translation models currently available is that they output isolated word tokens sequentially. Sign language follows its own grammar rules and sentence structures, i.e. the sign gestures do not have a 1-to-1 translation to their respective spoken counterparts. For example, the English sentence “I’m fine” would be represented in ASL using the signs: “Me, fine, me”. Due to this quirk of sign languages, the typical translation models completely fail to structure them into syntactically correct, grammatically fluid sentences.

5. PROJECT AIM & OBJECTIVES

Aim:

To engineer a comprehensive, real-time, bi-modal ASL translation ecosystem that utilizes spatial landmark optimization to remain computationally efficient on consumer-grade hardware.

Objectives:

- To implement local video stream acquisition via OpenCV and isolate regional hand boundaries.
- To extract spatial hand geometries as mathematical coordinate matrices using a MediaPipe framework, eliminating background visual noise.
- To deploy a dual-engine classification model: utilizing a Random Forest architecture for rapid, static finger-spelling recognition, and a CNN-LSTM hybrid network for temporal analysis of dynamic multi-frame gesture sequences.
- To integrate a Large Language Model (Gemini API) processing layer to map raw predicted word tokens into grammatically correct, context-aware English text.
- To synthesize the refined text output into high-quality auditory output via a Google Text-to-Speech (gTTS) engine.
- To design a centralized user dashboard to handle visual frame rendering, confidence monitoring, fallback verification prompts, and conversational history tracking.

6. LITERATURE REVIEW

The structural architecture of our proposed system is backed by two foundational research approaches:

Paper 1: [*Real-Time American Sign Language \(ASL\) Translator using OpenCV and MediaPipe \(IEEE, 2025\)*](#)

- **Core Approach:** This research develops a low-cost, scalable translation system by bypassing heavy deep learning frameworks. It maps hand gestures purely using structural coordinates and evaluates them via a Support Vector Machine (SVM) classifier.
- **Key Findings & Gaps:** The system achieves excellent computational latency, making it highly suitable for real-time local deployment. However, it displays performance drops under low-light conditions and is limited to simpler gesture subsets. The paper notes future goals as: training on a larger gesture dataset, integration of CNN for higher accuracy, implementation of increased accessibility features such as Text-to-Speech (TTS) modules and mobile/web release.

Paper 2: [*Hand Speak: An AI-Powered Real-Time System for Sign Language Recognition and Seamless Translation \(2025\)*](#)

- **Core Approach:** This paper addresses complex, continuous video streams by implementing a robust CNN framework optimized for live video gesture translation. It incorporates an NLP translation pipeline, speech synthesis, and real-time confidence scores.
- **Key Findings & Gaps:** To maintain real-time performance while handling heavy deep learning architectures, the system utilizes optimization techniques such as model pruning, quantization, and multi-threaded processing. This validates our strategy of using deep learning frameworks exclusively for complex dynamic movements while applying optimization techniques to keep execution lightweight.

7. PROPOSED METHODOLOGY / ARCHITECTURE

Functional Workflow:

1. **Frame Interception & Tracking:** The system captures live video via standard webcams using OpenCV. MediaPipe maps 21 key coordinate landmarks per hand, turning millions of raw pixels into lightweight spatial data vectors.
2. **Hybrid Inference Engine:** Static alphabet inputs are directed to a Scikit-Learn Random Forest model for immediate categorization. Dynamic multi-frame word tracking is routed to a TensorFlow/Keras-trained CNN pipeline.
3. **Linguistic & Syntactic Correction:** The generated predictive tokens are packaged sequentially and sent via API to a Gemini engine. The LLM handles conversational context mapping, correcting sentence syntax and managing conversational memory windows.

4. **UI and Dashboard:** The clean sentence outputs are displayed on a dashboard alongside real-time classification confidence metrics
5. **Accessibility:** The system will offer Text-to-Speech (TTS) support via Google TTS.
6. **Dataset:** The RF classifier will be trained on the ASL Alphabet dataset(Kaggle). Custom dynamic gestures sequences will be recorded for CNN training using MediaPipe Holistic keypoints.
7. **Evaluation:** System performance will be assessed during classification accuracy, F1 score, BLEU score for translation quality, latency benchmarks.

8. HARDWARE & SOFTWARE REQUIREMENTS

Software Requirements:

- **Programming Language:** Python 3.10+
- **Computer Vision Libraries:** OpenCV, MediaPipe
- **Machine Learning Frameworks:** Scikit-Learn (Random Forest), TensorFlow / Keras (CNN)
- **Natural Language Processing & Synthesis:** Gemini API, Google Text-to-Speech (gTTS)
- **User Interface Framework:** Streamlit

Hardware Requirements:

- Standard integrated or external USB Web Camera (30 FPS capability)
- Standard consumer-grade host processor (Core i5 or equivalent; specialized high-end cloud server GPUs are non-mandatory due to our landmark vector optimization).

9. EXPECTED OUTCOMES

- A fully functional, low-latency translation system capable of mapping static and dynamic hand movements into text.
- Significantly lower processing overhead compared to traditional raw pixel-based deep learning pipelines.
- High-context syntactic processing that produces conversationally sound, complete English translations instead of disconnected words.
- An intuitive, zero-install interactive UI web dashboard optimized for desktop and mobile layouts.
- BLEU score ≥ 0.75 for LLM translated sentences.
- More than 95% classification accuracy on the ASL alphabet under normal indoor lighting.